

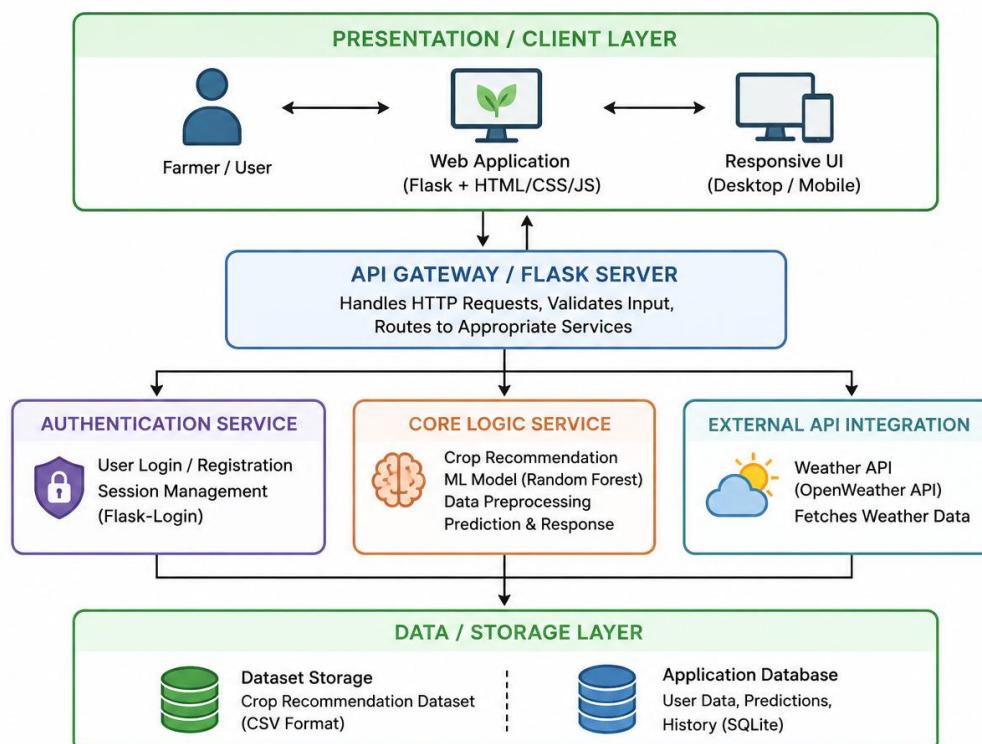
Solution Architecture

Date	05 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

SOLUTION ARCHITECTURE DIAGRAM



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	<i>Provides the user interface where farmers enter soil and weather details and view crop recommendations.</i>	<i>HTML, CSS, Flask (Frontend)</i>
API Gateway	<i>Receives user requests from the interface and forwards them to the machine learning prediction service.</i>	<i>Flask REST API, Python]</i>
Microservices	<i>Processes user input, performs data preprocessing, loads the trained ML model, and predicts the most suitable crop.</i>	<i>Python, Scikit-learn, Pandas, NumPy, Joblib</i>
Database	<i>Stores agricultural dataset, trained model files, prediction data, and application resources</i>	<i>CSV Dataset (Kaggle), Joblib (.pkl Model Files), File Storage</i>